# Ali Nikkhah

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## Summary

Backend and systems engineer with production experience building high-concurrency async APIs, distributed streaming infrastructure, and edge-optimized ML serving pipelines. At Digikala, engineered the async FastAPI serving layer handling 5M+ daily queries, with multi-tier Redis caching, vLLM PagedAttention, and Triton Inference Server for LLM deployment. At Advanced Analytics Australia, delivered distributed video analytics infrastructure on AXIS edge hardware using Kafka, TensorRT, and Kubernetes. Full ownership of the DevOps lifecycle: Docker, Kubernetes, Terraform, ArgoCD, and full observability stacks in production.

## Technical Skills

**Languages & Runtimes**: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
**Deep Learning & Core AI**: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
**Agentic AI & RAG**: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
**Vision & Audio**: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
**Data Engineering**: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
**MLOps & Serving**: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
**DevOps & Infrastructure**: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
**Databases & Monitoring**: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

## Professional Experience

**Digikala** — Tehran, Iran
*AI Engineering R&D Lead (Backend & Systems)* — *Jan 2025 – Aug 2025*

- Engineered the async **FastAPI** serving infrastructure handling over **5 million daily queries**; implemented multi-tier **Redis** caching and asynchronous request batching to fully isolate model execution latency from client-facing response times.
- Integrated **vLLM** with PagedAttention and **Triton Inference Server** for LLM and embedding model deployment, maximizing KV-cache utilization and GPU throughput under variable query burst patterns.
- Designed a Modular Control Pipeline (MCP) decoupling text ingestion, chunking, retrieval, and synthesis phases, enabling zero-downtime model hot-swapping and live index migrations.
- Built stateful multi-agent routing networks using LangGraph cyclic directed graphs to manage multi-turn session state, dynamic context evaluation, and automated query rewriting loops.
- Supervised Kubernetes deployment of multi-tenant model serving clusters, isolating vision and language model compute nodes for independent horizontal scaling and fault containment.

**Advanced Analytics Australia** — Remote Contract
*Backend & Edge Systems Engineer* — *Sep 2025 – Apr 2026*

- Built high-performance async Python streaming workers interfacing directly with AXIS network camera hardware, processing continuous live video feeds and routing them into a distributed **Apache Kafka** ingestion broker for fault-tolerant delivery under network spikes.
- Compiled and quantized vision networks into **TensorRT** and **TF Lite** runtimes served via Triton Inference Server, eliminating Python-level inference overhead on resource-constrained edge nodes.
- Designed sensor calibration routines and signal-to-noise monitoring to maintain stable detection thresholds across indoor and outdoor deployment sites.

**Turquoise Digital (Firouzeh Digital)** — Tehran, Iran
*Senior Data Engineer* — *Sep 2025 – Mar 2026*

- Developed async backend infrastructure in Python/FastAPI supporting high-frequency portfolio projection requests and automated asset trading execution; decoupled ingestion from core relational databases via multi-tier Redis caching and **Apache Kafka** message queuing.
- Maintained full system reliability and observability across microservices using **Prometheus**, **Grafana**, and the ELK log aggregation stack.
- Orchestrated complex **Airflow** DAGs and deployed **PeerDB** CDC replication clusters to continuously mirror transactional databases into OLAP engines with sub-minute latency.

**HomaCloud** Tehran, Iran
*MLOps Engineer* *Aug 2021 – Feb 2022*

- Automated end-to-end model deployment pipelines with Docker, **ArgoCD**, and GitHub Actions, reducing release cycles to under 10 minutes; managed Kubernetes cluster orchestration and IaC via Terraform.
- Built low-latency model inference APIs with FastAPI and TensorFlow Serving to handle concurrent traffic spikes; orchestrated distributed workloads with Ray and MLflow.

## Selected Projects & Research

**Distributed Intelligent Surveillance Infrastructure** Advanced Analytics Australia
*Core Backend & Edge Systems Architect* *2025 – 2026*

- Designed the full backend architecture for a multi-site industrial safety compliance system: async Python streaming workers feed a Kafka broker, which distributes camera frames to TensorRT/TF Lite inference containers served via Triton, with detection events written to a PostgreSQL audit store.
- Implemented dynamic sensor calibration and continuous SNR monitoring to maintain stable detection thresholds across variable lighting and exposure conditions at outdoor and indoor sites.

**Enterprise Agentic AI Serving Infrastructure** Digikala
*Lead Backend Engineer* *2024 – 2025*

- Architected the complete serving layer for a multi-agent RAG system at 5M+ daily queries: FastAPI front-end, vLLM/Triton model runtime with PagedAttention, multi-tier Redis cache, and LangGraph session-state management — all deployed on a Kubernetes multi-tenant cluster.

## Research Highlights

**University of British Columbia** Remote Research Collaboration
*Research Assistant – Vision-Language Medical Imaging* *Feb 2024 – Nov 2025*

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

**L3S Research Center (Leibniz University Hannover)** Remote Research Collaboration
*Research Assistant – Video Motion Classification* *Apr 2023 – Feb 2024*

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

## Education

**Sharif University of Technology** Tehran, Iran
*B.Sc. Electrical Engineering – Digital Systems* *GPA: 3.65 overall, 3.91 major* *Sept 2020 – Jul 2024*